

Department of the Interior
U.S. Geological Survey

Landsat Flight Operations (LFO) Project System Configuration Management (CM) Policy

Version 1.0

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Executive Summary

This document establishes the practices and processes for the Landsat Flight Operations (LFO) Project and its subordinate structures. A Configuration Management (CM) Policy governs Configuration Item (CI) change management throughout the lifecycle of projects under the Land Satellites Data System (LSDS) Program. This CM Policy details the policies, processes, and procedures that establish and maintain the baseline integrity of the LFO Project, including roles and responsibilities, Change Requests (CRs), and which Configuration Control Boards (CCBs) govern configuration management.

This document is a Configuration Management (CM)-controlled document for the Landsat Flight Operations (LFO) Project. Changes to this document require prior approval of the Landsat Ground System (GS) Configuration Control Board (CCB). Please submit proposed changes to this document, as well as supportive material justifying the changes, to the Landsat Mission Manager (LMM).

Questions or comments concerning this document should be addressed to:

- Landsat Flight Operations Project Manager
- Landsat Mission Manager

Document History

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Section 1 Purpose and Scope of Configuration Management

1.1 Background

Landsat is an Earth observation satellite program formulated, implemented, and operated by the National Aeronautics and Space Administration (NASA) and the Department of the Interior (DOI) U.S. Geological Survey (USGS). Landsat's ongoing record of data focuses on medium-resolution remote sensing of Earth's land surfaces.

The goal of Landsat is to continue the collection, archive, and distribution of multispectral imagery affording global, synoptic, and repetitive coverage of land surfaces at a scale where natural and human-induced changes can be detected, differentiated, characterized, and monitored over time. The mission's programmatic goals are stated in the United States Code, Title 15 Chapter 82 "Land Remote Sensing Policy" (derived from the Land Remote Sensing Policy Act of 1992). This policy requires the Landsat program to provide data into the future that are sufficiently consistent with previous Landsat data, allowing the detection and quantitative characterization of changes in or on the Earth's surface. The highly successful Landsat series of missions has provided satellite coverage since 1972. The data from these missions constitute the longest continuous record of Earth's surface as seen from space.

1.2 Overview

Landsat Flight Operations (LFO) personnel oversee activities related to the Landsat observatories and ground system. The observatories include the on-orbit integrated spacecraft and science instruments required for the acquisition and downlink of Landsat science data. The ground system includes the Mission Operations Centers (MOCs), their backup MOCs (bMOC), and the infrastructure to support their functions.

1.3 Purpose and Scope

The purpose of this document is to establish and maintain the integrity of Land Satellites Data System (LSDS) Program documentation, hardware, and software within the scope of the Landsat Flight Operations (LFO) Project and its subordinate structures.

The scope of this document includes the structure for Configuration Management (CM) for LFO ground- and space-based assets, and the identification and control processes for the physical, virtual, hardware, software and logical Configured Items (CIs), as well as the process of verifying and auditing baselined CIs using varying types of Change Requests (CRs).

1.4 Document Organization

This document contains the following sections:

- Section 1 provides an overview of the document purpose and scope
- Section 2 outlines the components of configuration control

- Section 3 details CM roles and responsibilities
- Section 4 defines Change Requests (CRs)
- Section 5 outlines the CM workflow
- Appendix A provides a list of acronyms

1.5 References

1.5.1 Reference Documents

The documents in Table 1-1 provide additional background and context for this document.

Document Number	Version	Document Title
LSDS-447	10.0	Land Satellites Data System (LSDS) Configuration Control Board (CCB) Charter
LSDS-448	11.0	Land Satellites Data System (LSDS) Change Management Plan (CMP)
LSDS-459	6.0	Land Satellites Data System (LSDS) Hardware Configuration Management Plan (HCMP)
LSDS-465	6.0	Land Satellites Data System (LSDS) Software Configuration Management Plan (SCMP)
LSDS-485	12.0	Land Satellites Data System (LSDS) Document Management Plan (DMP)
LSDS-1346	2.0	Landsat Multi-Satellite Operations Center (LMOC) Operations Configuration Control Board (O-CCB) Charter
LSDS-1875	2.0	Landsat Ground System (GS) Configuration Control Board (CCB) Charter
LSDS-1908	2.0	Landsat Ground System (GS) Environments (ENV) Configuration Control Board (CCB) Charter

Table 1-1. Reference Documents

1.6 Configured Item

CIs within configuration management (CM) require formal change control. Formal change control requires Configuration Control Board (CCB) management using CRs associated with the governing CCB. The following list identifies example CIs applicable to ground systems and operational Observatory systems. This list is not comprehensive and additional CIs can be considered for CM control at the discretion of any Configuration Control Board (CCB) Chair.

- Documents
 - Build Guides and User's Guides
 - Software Build Guide (SBG)
 - User's Guide (UG)
 - Charters
 - Configuration Control Board (CCB)

- Integrated Project Team (IPT)
- Contingency Operating Procedure (COP)
- Contract Deliverable Requirements List (CDRLs)
- Data Format Control Book (DFCB)
- Description Documents
 - Algorithm Description Document (ADD)
 - Database Description Document (DBDD)
 - Version Description Document (VDD)
- Design Documents
 - Architecture
 - System
 - Software
- Diagrams / Drawings
 - Specification Tree
 - Rack Hardware connectivity diagrams
 - Network connectivity diagrams
- Interfaces
 - Interface Control Document (ICD)
 - Interface Specification Document (ISD)
- Program and Project Management Plans
 - Program and Project Specific Management Plans
 - Governance Plan
- Operational Concept Document (OCD)
- Policies and Standards
- Requirements Documents
 - System Requirements Document (SRD)
 - Software Requirements Specification (SRS)
- Scope
 - Contract Work Breakdown Structure (CWBS) and CWBS Dictionary
 - Program Work Breakdown Structure (PgWBS) and Dictionary
 - Project Work Breakdown Structure (PWBS)
- Standard Operating Procedures (SOP)
- Statement of Work (SOW)
- Templates
- Test Plans (Detailed with steps)
- Software / Hardware
 - Configuration Files
 - Development Project Build Packages
 - Development Project Install Packages
 - Firmware
 - Operations (Ops) Workstations
 - Operational Project Build Packages
 - Operational Project Install Packages
 - Configurable Item Identification List (CIIL)
 - SCRs
 - SpCRs

- Ops Products
 - Procs
 - Pages
 - Trending Batch files
 - Reports
 - Command load Building files
 - Load checking software
 - Activity Definitions
 - LCR Definition
 - Alarm Response Database
 - Observatory Memory Ground Reference Image(s)
 - Vendor provided users manuals
 - Vendor provided trainings
 - Automation scripts
 - Operational scripts
- Radio Frequency (RF) Equipment / Antenna
- Servers, Storage, and Configuration
- **System interface configuration control**
- System Release (software / code tagging)

1.7 Non-Configured Item

Non-CIs within configuration management do not require formal change control. Non-CIs can be controlled and managed through the change management process; non-CIs are not presented at the CCB. While the following are typically non-configured, projects may choose to configure certain non-CIs to meet specific project requirements. The following list identifies example non-CIs. This list is not comprehensive and additional non-CIs can be added at the discretion of any Configuration Control Board (CCB) Chair.

- Documents
 - Contract and Financial Reports
 - Monthly Invoices
 - Contractor Monthly Status Report (MSR)
 - Government Monthly Status Reports (MSR)
 - Commercial-Off-The-Shelf (COTS) / Government-Off-The-Shelf (GOTS) Guides
 - Diagrams / Drawings
 - Architecture
 - Entity Relationship Diagram (ERD)
 - Forms
 - Inventory Report (Nlyte)
 - Lists
 - Acronyms
 - Lexicon
 - Master Document List (MDL)
 - Master Software List (MSL)
 - Master Hardware List (MHL)

- Master Build/Install List
 - Property List
- Milestone Reviews
 - Presentations / Materials
- Program and Project Level Notification Milestones
- Procurement Reports
- Schedule
 - Progress Reports
 - Snapshots
- Task Order (TO), Statement of Objectives (SOO)
- Test Artifacts
 - Cases
 - Discrepancies
 - Reports
- Test Plans (no steps)
- Trade Studies and White Papers
- Engineering Reports
- Anomaly Reports
- Software / Hardware
 - COTS / GOTS
 - Database Entries / Content
 - Personal Computers (PCs) and PC Software
 - Tools / Converters / Utilities
 - Peripheral Devices (Printers, plotters, scanners, USBs)
- Data/Files
 - Operations artifacts
 - Command Loads
 - Stored engineering or image data
 - Reports (in general)
 - SCRs/SpCRs

Section 2 Configuration Control

2.1 Configuration Control Components

The CCBs and corresponding CRs are the components that facilitate CM. The CCB structure is a multilevel model that allows the Landsat Flight Operations (LFO) Project and adjacent programs within the LSDS Program to address CRs at the appropriate level. The structure also allows CRs to flow through a formal approval process to control and implement change within each level. The CCB manages the controlled CI baseline, coordinates change implementation, and approves the controlled documentation associated with the CCB. The appropriate CCB Charter identifies the CCB purpose, scope, structure, and membership. The primary CCBs used by LFO are the

- LSDS CCB
- OPRB/OCCB (Landsat Multi-Satellite Operations Center (LMOC) Operations Configuration Control Board)
- Landsat Ground System (GS) CCB

Please see reference documents LSDS-1346 and LSDS-1875 for more detailed information on their CI scope and governing authority.

CRs are submitted through the appropriate channels as defined in the charters for the aforementioned CCBs. These CRs provide traceability for the addition of new items and existing item modification(s) for CIs under a CCB's control. Each CI is administered by one CCB that has ownership and authority over the CI. All CIs associated with a CCB are documented within the appropriate associated master list, including their associated configuration management controlled repository.

Each CCB should have an associated CM-controlled repository that contains the CM managed baseline of items (CI and/or Non-CI) governed by the CCB. CCBs can share repositories. The CM repositories and their contents should be defined in the CCB Charter. Please see LSDS-1346 and or LSDS-1875 for more information and examples.

If there is no stipulation in their respective SOWs requiring an internal contractor-led CCB, the LFO contractors are encouraged to define their own contractor-led CCBs which report up to the LFO CCBs. *The LFO Project reserves the right to delegate CI and Non-CI CM authority down to the contractor-led CCBs for items described Sections 1.6 and 1.7, at the discretion of the Landsat Mission Manager (LMM) or appropriate Flight Systems Manager (FSM).*

Section 3 CM Roles and Responsibilities

3.1 Configuration Management Lead

The Configuration Management (CM) Lead is responsible for LFO Project and/or subordinate CCB CM functions. The CM Lead responsibilities include the following:

- Maintain the appropriate CCB Charter(s)
- Convene CCB meetings on a regular basis
- Implement the procedures outlined in this CM Policy and associated CCB charters
- Ensure the CM processes and guidelines are followed
- Ensure personnel involved in the CM processes are properly trained
- Establish CM policies with CCB approval
- Establish an agenda prior to the CCB meeting
- Disseminate the agenda to CCB members prior to scheduled CCB meetings
- Facilitate the CCB meeting
- Create and disseminate CCB meeting minutes to CCB members
- Assist with CR submissions
- Ensure completion of CM CI changes associated w/ approved CRs
- Primary Steward of the CM controlled repository(ies)
- Audit CIs for applicability based on their effective dates (particularly for documents)

3.2 Software Configuration Manager

The Software Configuration Manager responsibilities include the following:

- Maintain the software configuration management plans applicable to each project
- Identify nominal baseline software operational configuration(s)
- Provide configuration control
- Perform baseline audits
- Manage software builds
- Manage software releases
- Design, implement, manage, and maintain the Software Configuration Management (SCM) infrastructure

3.3 Hardware Configuration Manager

The Hardware Configuration Manager responsibilities include the following:

- Maintain the hardware configuration management plans applicable to each project
- Identify nominal baseline hardware operational configuration(s)
- Provide configuration control
- Perform baseline audits
- Manage hardware installation, warranty exchange, and refresh

3.4 CCB Chair

Each CCB charter identifies the CCB's voting membership (Chairs), as well as the supporting members needed for a quorum. Each list does not preclude additional or ad hoc attendees at meetings or review of the operations CIs. The charter members are encouraged to consult with other individuals outside the routine meeting time and invite Subject Matter Experts (SMEs) to address specific agenda topics, as necessary. To fulfill a quorum, a designated representative may be assigned.

The CCB Chair(s) at each CCB level is responsible for the following:

- Convene special or additional CCBs as needed
- Attend and chair CCB meetings
- Delegate authority in the event of their absence or contact the CM Lead to cancel the CCB meeting
- Ensure each CR receives appropriate CCB member review
- Review all CCB member recommendations and comments
- Approve, conditionally approve, or reject CRs within the CCB scope

Section 4 Change Requests / *Pull Requests*

4.1 CRs

CRs are the path to initiate a desired change. CRs are submitted through the appropriate channels as defined by each CCB, at times with CM Lead assistance. Approved CRs result in the desired change being implemented/deployed/saved in the CM-controlled repository of the governing CCB.

4.1.1 Types of CRs

CRs consist of three types, which include documentation, software, and hardware. For details, please reference LSDS-485 Land Satellites Data System (LSDS) Document Management Plan (DMP), LSDS-459 Land Satellites Data System (LSDS) Hardware Configuration Management Plan (HCMP), and LSDS-465 Land Satellites Data System (LSDS) Software Configuration Management Plan (SCMP).

4.1.2 CR Priority Definitions

CRs have three prioritization levels. The prioritization level identifies the CR urgency level required to pass through the CCB process. All CRs are assigned a “Routine,” “Urgent,” or “Emergency” priority level; however, all CRs default to “Routine.” Upon entering a CR into the CM Tool, the submitter identifies the CR priority based upon the appropriate criteria. The CCB Chair has the discretion to authorize an electronic CCB at any priority level.

CRs are prioritized as follows:

- Routine: The CR is deemed acceptable to address or resolve at the next scheduled CCB meeting.
- Urgent: Form, fit, function, performance, schedule, and/or cost will be severely impacted if the CR is delayed. CRs shall receive CCB disposition within five days of CR submission. Urgent CR types may be handled at the next routine CCB, an out-of-cycle CCB meeting, or an electronic CCB depending on the CCB type that fits the urgency.
- Emergency: The condition is deemed such that failure to implement this change seriously compromises the mission, when hazardous conditions exist, or may be necessary due to an unexpected near-term deadline for a document / CI update. Emergency CRs receive disposition within 24 hours of receipt. The CM Lead notifies all CCB members via email that an emergency CR exists and must be addressed immediately. Emergency CR types are typically completed as an electronic CCB.

Changing the priority of a CR is at the discretion of the CM Lead, based upon initial analysis after submission. The CCB may also change the priority of a CR at review.

Section 5 Configuration Management Overview

All CRs are submitted through the appropriate channels as defined by each CCB. Documentation, software, and hardware CIs follow the same CR lifecycle with the exception of the CCB approval state. Documentation is submitted to the CCB for approval once the work is complete, whereas software and hardware are submitted to the CCB for approval prior to beginning the work and return to the CCB for closure notification prior to closing the CR.

LSDS-485 Land Satellites Data System (LSDS) Document Management Plan (DMP) identifies the documentation lifecycle and process. LSDS-465 Land Satellites Data System (LSDS) Software Configuration Management Plan (SCMP) identifies the software CR lifecycle and process. LSDS-459 Land Satellites Data System (LSDS) Hardware Configuration Management Plan (HCMP) identifies the hardware CR lifecycle and process.

Figure 5-1 provides an overview of configuration management for all CI types.

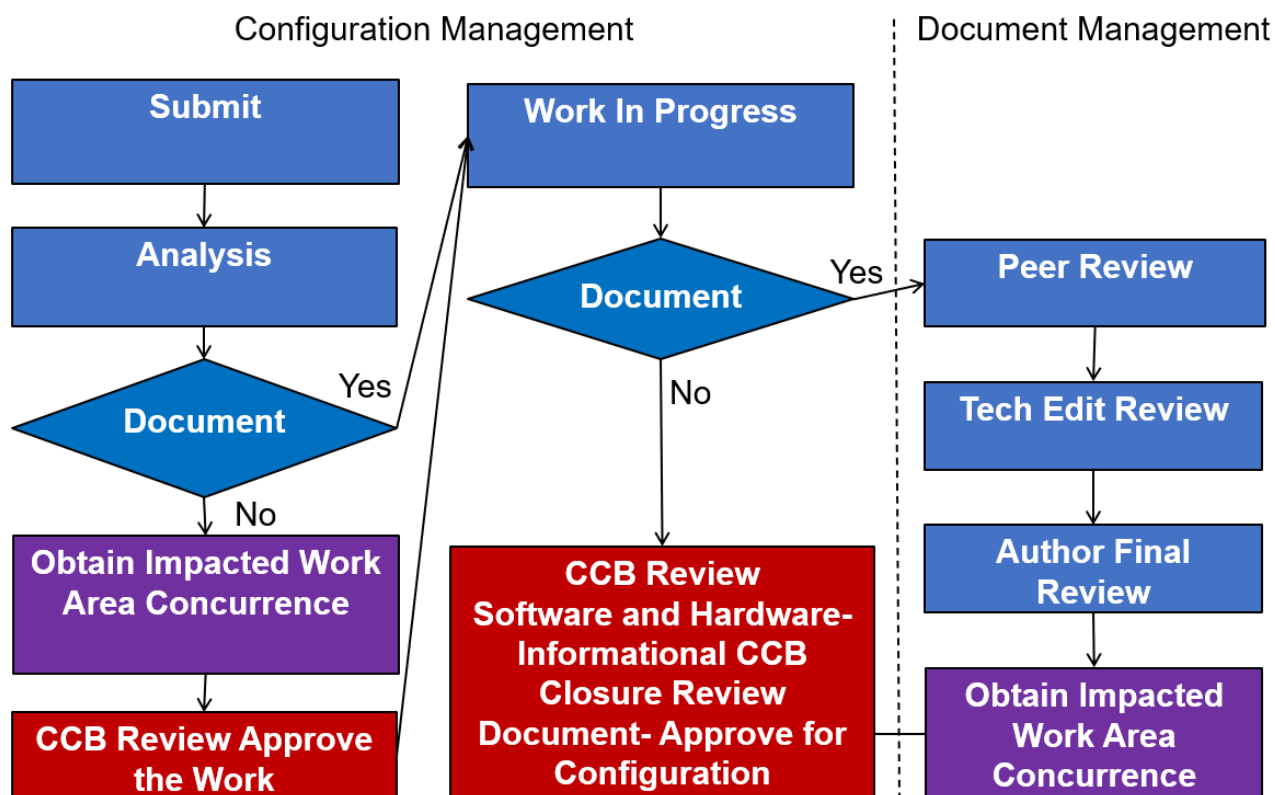


Figure 5-1. Configuration Management Overview

Each CI and/or Non-CI housed in a CM-controlled repository should be assigned effectivity duration upon their CR's approval. The CM Lead should monitor the effectiveness dates of all CIs within the repositories under their purview, and all CIs that

have effective dates coming to a close should be reviewed for applicability. This can be done by re-submitting the CR to the CCB, but that process/decision is ultimately up to the CCB Chair. Please see the appropriate CCB Charter for any additional information.

Appendix A Acronyms

Please see <https://www.usgs.gov/landsat-missions/landsat-acronyms> for a complete list of acronyms.

Acronym	Definition
ADD	Algorithm Description Document
CCB	Configuration Control Board
CDRL	Contract Deliverable Requirements List
CI	Configuration Item
CIIL	Configurable Item Identification List
CM	Configuration Management
CMP	Change Management Plan
COP	Contingency Operating Procedure
COTS	Commercial-Off-The-Shelf
CR	Change Request
CWBS	Contract Work Breakdown Structure
DBDD	Database Description Document
DFCB	Data Format Control Book
DMP	Document Management Plan
DOI	Department of the Interior
ERD	Entity Relationship Diagram
FSM	Flight Systems Manager
GOTS	Government-Off-The-Shelf
GS	Ground System
HCMP	Hardware Configuration Management Plan
ICD	Interface Control Document
IPT	Integrated Project Team
ISD	Interface Specification Document
LFO	Landsat Flight Operations Project
LMOC	Landsat Multi-Satellite Operations Center
LMO	Landsat Mission Operations
LMM	Landsat Mission Manager
LSDS	Land Satellites Data System
MDL	Master Document List
MHL	Master Hardware List
MMO	Landsat Mission Management Office
MOC	Mission Operations Center
MSL	Master Software List
MSR	Monthly Status Report
NASA	National Aeronautics and Space Administration
O-CCB	Operations Configuration Control Board
OCD	Operations Concept Document
Ops	Operations
PgWBS	Program Work Breakdown Structure

Acronym	Definition
RF	Radio Frequency
SBG	Software Build Guide
SCMP	Software Configuration Management Plan
SCM	Software Configuration Management
SOO	Statement of Objectives
SOP	Standard Operating Procedures
SOW	Statement of Work
SME	Subject Matter Expert
SRD	System Requirements Document
SRS	System Requirements Specification
TO	Task Order
UG	User Guide
USGS	United States Geological Survey
VDD	Version Description Document